Livestock Sector Forecasting in Kenya: A Cohort-Based Quarterly Analysis on Population, Production, and Feed Demand

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Abstract

Kenya's livestock sector contributes 42% of agricultural GDP and 12% of national GDP, yet quantitative planning remains constrained by the absence of reproducible, data-driven analytical models. This study presents a cohort-based quarterly forecasting framework spanning nine value chains: dairy cattle, beef cattle, camels, goats, sheep, poultry, pigs, rabbits, and apiculture, initialized from the 2019 KNBS livestock census with projections from 2020 to 2041. The model applies a quarterly transition sequence covering mortality, culling, offtake, age-based maturation, and reproduction across all cohorts. Production modules translate herd and flock dynamics into species-specific output metrics, while a carrying capacity constraint — derived from satellite-based vegetation indices and adjusted by a utilization factor — regulates growth against forage availability. A feed demand module estimates dry matter requirements by species and agro-ecological zone. Projected outputs indicate a CAGR of 3.6-6% across all value chains. Beyond forecasting, the framework exposes productivity bottlenecks and structural inefficiencies, including feed deficits, low offtake rates, and suboptimal reproductive performance, that aggregate assessments typically obscure. The model provides an analytical basis for food security planning, targeted infrastructure investment, and evidence-based livestock policy.

Keywords: Livestock model forecasting, carrying capacity, biological parameters

1 Introduction

In most African countries, the livestock sector supports the livelihoods of millions of rural households and plays an important role in poverty reduction. In livestock-rich countries such as Kenya, the sector contributes substantially to the national economy while providing income, employment, and food security across both rural and urban populations. Beyond primary production, livestock also supports employment throughout value chains, including dairy, meat, and leather processing industries. The livestock sector plays a major role in Kenya's food system, contributing approximately 12% of the country's Gross Domestic Product (GDP), 40% of agricultural GDP, and 22% of the food system GDP [1].

Kenya's population is projected to reach approximately 96 million by 2050, with nearly half of the population residing in urban areas. Consequently, demand for animal-source foods is expected to increase substantially owing to population growth, urbanization, rising incomes, and changing dietary preferences [1]. Meeting this growing demand will require sustained improvements in livestock productivity, efficient management of feed and water resources, and evidence-based planning to support the sustainable development of the livestock sector. Reliable forecasting frameworks that project livestock populations, production outputs, feed demand, and resource requirements are therefore essential for evidence-based planning, strategic investment, and sustainable livestock sector development.

Statistical time-series models are commonly used to forecast livestock populations using historical data. These methods estimate future livestock numbers based on past population trends and have been applied to support livestock planning and management. [1] used autoregressive and probabilistic time-series models to forecast pig populations, while Ozbek compared autoregressive integrated moving average (ARIMA) and artificial neural network (ANN) models for forecasting cattle populations in Türkiye. Similarly, [1] applied exponential smoothing to forecast beef cattle populations in Indonesia. Although these methods have demonstrated good predictive performance, they rely primarily on historical trends and do not explicitly represent biological processes such as reproduction, mortality, maturation, culling, and offtake. Consequently, their ability to evaluate the effects of biological parameters and management interventions on future livestock population dynamics is limited.

In contrast to statistical forecasting approaches, biological simulation models explicitly represent the demographic processes governing livestock populations. Animals are grouped into cohorts according to age, sex, or production stage, with transitions driven by biological processes such as reproduction, maturation, mortality, culling, and replacement. These models offer a biologically realistic representation of herd dynamics and enable the evaluation of alternative management strategies. [1] developed an age- and sex-structured cattle population model that incorporates management practices and animal movements to simulate herd dynamics. Such models serve as valuable decision-support tools by linking biological processes to herd structure and productivity, they are generally developed for specific livestock species or production systems and do not explicitly integrate livestock population dynamics, production, environmental carrying capacity, and feed demand within a unified forecasting framework. Consequently, there remains a lack of reproducible analytical frameworks that simultaneously integrate livestock population dynamics, production, environmental carrying capacity, and feed demand to support long-term livestock sector planning.

To address these limitations, this study develops a cohort-based quarterly forecasting framework for Kenya's nine principal livestock value chains. The framework explicitly models biological cohort transitions through reproduction, mortality, maturation, culling, and offtake while linking livestock population dynamics to species-specific production, environmental carrying capacity, and feed demand across agro-ecological zones. By integrating these components within a single modelling framework, the proposed approach provides a comprehensive decision-support tool for forecasting livestock populations, production, and resource requirements, thereby supporting evidence-based livestock planning and policy development in Kenya.

2 Literature Review

Write related work (literature review) here.

3 Methodology

3.1 Overview

Eight value chains (both ruminant and non-ruminant animals) were selected on the basis of their contribution to Kenya's agricultural GDP.[1] For each chain, forecasting covers three domains: **population dynamics**, **production volumes**, and **feed demand**. The baseline year is 2019; the projection horizon runs from 2020 to 2041, with 2020–2025 retained for model validation against observed values.

A first-principles modelling approach was adopted. This choice reflects (i) the transparency and auditability it affords where each intermediate quantity is directly traceable to its inputs, and (ii) the heterogeneous, multi-source nature of the available data, which precludes the large homogeneous datasets required by machine-learning methods.

Parameters were compiled from peer-reviewed literature, government publications, and sector reports, then validated by livestock experts from the Ministry, associations, and farmers. Accepted values are listed in Appendix 1.[2]

The model operates on a quarterly time step. For each quarter, the three domains are computed in sequence.

3.2 Population Model Equations

Population is tracked across a defined cohort structure specific to each value chain. Within each quarter, the following processes are applied in order where relevant: mortality, culling, offtake, transition to the next age cohort, and births. The starting population for each cohort is updated by sequentially applying the corresponding rates, with inflows from younger cohorts and new births added at the end of the period.

[1] Citation required, get from Natasha
[2] Summarize parameters

3.2.1 Beef Cattle

$$P(t+1) = \big[P(t)\,(1-m_q)(1-k_q)(1-o_q)\big] + G(t) - E(t) + B(t) \tag{1}$$

$P(t)$	Starting pop.	m_q	Mortality rate (qtr.)
$P(t+1)$	Ending pop.	k_q	Culling rate
$G(t)$	Inflow (young cohort)	o_q	Offtake rate (steers)
$E(t)$	Outflow to next cohort	$B(t)$	New births (calves)

3.2.2 Goats

$$P(t+1) = \big[P(t)\,(1-m_q)(1-k_q)\big] + G(t) - E(t) + B(t) \tag{2}$$

Variable definitions follow the same convention as beef cattle; the offtake term o_q does not apply.

3.2.3 Camels

$$P(t+1) = P(t) - D(t) - C(t) - O(t) + B(t) \tag{3}$$

$D(t)$	Natural mortality	$C(t)$	Culling offtake
$O(t)$	Commercial offtake	$B(t)$	Births in period t

3.2.4 Poultry

Three sub-sectors are modelled separately. T_{bro} denotes annual broiler throughput and S_r the rearing survival rate.

$$P_{\text{ind}}(t+1) = P_{\text{ind}}(t) - E_{\text{ind}}(t) + R_{\text{ind}}(t) + A(t) \tag{4}$$

$$P_{\text{lay}}(t+1) = P_{\text{lay}}(t) - E_{\text{lay}}(t) + \text{DOC}(t) \times S_r \tag{5}$$

$$T_{\text{bro}}(t+1) = T_{\text{bro}}(t) + A(t) \tag{6}$$

The sector addition term $A(t)$ and its sigmoid growth modifier $g(t)$ are:

$$A(t) = P(t)\,(r_{\text{ent}} + r_{\text{exp}})\,g(t) \tag{7}$$

$$g(t) = g_{\min} + (g_{\max} - g_{\min})\,\exp\!\left(-\kappa^2\,\frac{(t-t_0)^2}{2}\right) \tag{8}$$

3.2.5 Apiculture

$$H(t+1) = H(t)\,(1-\delta(t))(1-\rho(t)) + S(t) \tag{9}$$

$\delta(t)$	Seasonal colony loss	$\rho(t)$	Queen replacement rate
$S(t)$	New colonies (swarming)		

3.2.6 Pigs

$$P(t+1) = P(t) - D(t) - C(t) - O(t) + B(t) + R(t) \tag{10}$$

where $R(t)$ represents replacement additions to the breeding herd. Variable definitions for $D(t)$, $C(t)$, $O(t)$, $B(t)$ follow the camel convention.

3.2.7 Rabbits

Rabbits follow the same population equation as pigs (including $R(t)$).

3.3 Production Model Equations

Production volumes are derived from the population state at the end of each quarter. Each value chain has a distinct production equation reflecting the commodity it yields, that is meat, milk, eggs, honey, or wax, as detailed in Sections II-B through II-C. [3] For ruminants, production is computed separately for each cohort and then aggregated. For poultry, the three sub-sectors are modelled independently. Apiculture production is calculated at the hive level and then scaled by the number of hives.

3.3.1 Beef Cattle

$$\text{Beef (kg)} = \sum_c N_c \times \text{LW}_c \times \text{DP}_c \tag{11}$$

where $c \in \{\text{steers, cull cows, cull bulls}\}$, LW is live weight (kg) and DP is dressing percentage.

3.3.2 Goats

$$\text{Meat (kg)} = (N_{\text{removed}} + N_{\text{excess bucks}}) \times \text{LW} \times \text{DP} \tag{12}$$

$$\text{Milk (kg)} = N_{\text{does}} \times f_L \times Y_d \times D_L \tag{13}$$

N_{does}	Surviving does
f_L, Y_d, D_L	Lactating fraction; daily yield (kg); lactation days
$N_{\text{excess bucks}}$	Bucks above 4% herd cap, removed for slaughter

3.3.3 Camels

$$\text{Milk (kg yr}^{-1}) = N_{\text{adult females}} \times f_L \times Y_d \times 365 \tag{14}$$

$$\text{Meat (kg)} = \sum_c N_c \times \text{LW}_c \times \text{DP}_c \tag{15}$$

where $c \in \{\text{juv. offtake, adult offtake, cull adults}\}$.

3.3.4 Poultry

$$\text{Eggs}_{\text{ind}}(t) = P(t) \times f_{\text{fem}} \times Y_e \tag{16}$$

$$\text{Eggs}_{\text{lay}}(t) = P(t) \times \eta_{\text{HD}} \times 365 \tag{17}$$

$$\text{Meat (kg)} = T_{\text{bro}} \times \text{LW} \times \text{DP} \tag{18}$$

[3] Update the section names

where f_{fem} is female share, Y_e is eggs per hen per year, and η_{HD} is hen-day production rate.

3.3.5 Apiculture

$$\text{Honey}_{h,q} = H_h \cdot \frac{Y_h}{4} \cdot \bar{s} \cdot (1 - \varphi) \cdot \Omega(t) \cdot \varepsilon_h \tag{19}$$

$$\text{Wax}_{h,q} = \text{Honey}_{h,q} \times \omega_h \tag{20}$$

$$\text{Nectar}_{h,q} = H_h \cdot \frac{N_h}{4} \cdot \bar{s} \cdot \phi_q \cdot \nu_q \cdot \Omega(t) \tag{21}$$

Y_h, $\bar{s}$	Annual yield/hive (kg); effective colony strength
φ	Forage competition penalty (0–0.20)
ε_h	Hive-type efficiency factor
$\Omega(t)$	Technology/training improvement (sigmoid)
ω_h	Wax-to-honey ratio
ϕ_q, ν_q	Quarterly forage and nectar seasonality factors

3.3.6 Pigs and Rabbits

$$\text{Meat} = N_{\text{slaughtered}} \times \text{LW} \times \text{DP} \tag{22}$$

3.4 Feed Model Equations

Feed demand is estimated using dry matter intake (DMI) for all value chains. A DMI-based approach was adopted due to limited availability of detailed ration composition data across most value chains; it provides a consistent and tractable basis for estimating feed requirements from population and live weight alone. The specific formulations vary by value chain and are presented in Section II-D.

3.4.1 Beef Cattle, Pigs, and Rabbits

All three value chains use the same daily dry matter intake (DMI) formulation:

$$\text{DMI} = P(t) \times \text{LW} \times r_{\text{DMI}} \times 365 \tag{23}$$

where r_{DMI} is daily dry matter intake as a proportion of body weight.

3.4.2 Goats

Feed is summed across cohorts (kids, weaners, growers, does, bucks):

$$\text{Feed}(t) = \sum_{\text{cohort}} N_c \times \text{BW}_c \times r_{\text{DMI},c} \times D \tag{24}$$

where D is the number of days in the period.

3.4.3 Camels

Feed is accumulated quarterly, with intake weighted by average live weight over the quarter:

$$\text{Annual feed} = \sum_{q=1}^{4} \text{Feed}_q \tag{25}$$

$$\text{Feed}_q = \sum_{\text{cohort}} N_c \times \left(\frac{\text{BW}_{0,c} + \text{ADG}_c \times 90}{2} \right) \times r_{\text{DMI},c} \times 90 \tag{26}$$

where ADG is average daily gain ($\text{kg}\,\text{d}^{-1}$).

3.4.4 Poultry

Each sub-sector has a distinct feed equation reflecting its production cycle structure:

$$\begin{aligned} F_{\text{ind}}(t) &= \big[R(t)(f_{\text{chk}}\, d_{\text{chk}} + f_{\text{gr}}\, d_{\text{gr}}) \\ &\quad + P(t)\, f_{\text{ad}}\, 365\big](1-\sigma)^t \end{aligned} \tag{27}$$

$$\begin{aligned} F_{\text{lay}}(t) &= \text{DOC}(t)(f_{\text{chk}}\, d_{\text{chk}} + f_{\text{pul}}\, d_{\text{pul}}) \\ &\quad + P(t)\, f_{\text{hen}}\, 365 \end{aligned} \tag{28}$$

$$F_{\text{bro}} = N_{\text{birds}} \sum_{\text{phase}} f_{\text{phase}} \times d_{\text{phase}} \times C_{\text{yr}} \tag{29}$$

where $f_{(\cdot)}$ is phase-specific intake ($\text{kg}\,\text{d}^{-1}$), $d_{(\cdot)}$ is phase duration (days), σ is the scavenging offset, and C_{yr} is production cycles per year.

4 Results

All projections run from a 2019 baseline to 2041, with 2020–2025 serving as the validation window (shaded in all figures). Compound annual growth rates (CAGRs) are reported over the full 2019–2041 horizon unless noted otherwise.

4.1 Ruminant Value Chains

4.1.1 Population

Cattle dominate the ruminant sector by absolute numbers. Beef cattle grow from 16.6 million in 2019 to 67.8 million by 2041 (CAGR 6.59%), while dairy cattle expand from 4.5 million to 13.1 million (CAGR 4.93%). Goats are the most numerous ruminant throughout the horizon, rising from 34.7 million to 121.3 million (CAGR 5.86%). Camels grow steadily from 4.7 million to 14.3 million (CAGR 5.15%), and sheep show the most modest growth, from 27.4 million to 55.7 million (CAGR 3.27%). Population trajectories are shown in Fig. 1.

4.1.2 Production

Milk production is led by dairy cattle (3.1 billion kg in 2019 to 9.3 billion kg by 2041; CAGR 5.12%), followed by camels (0.99 billion kg to 5.6 billion kg; CAGR 8.23%) and

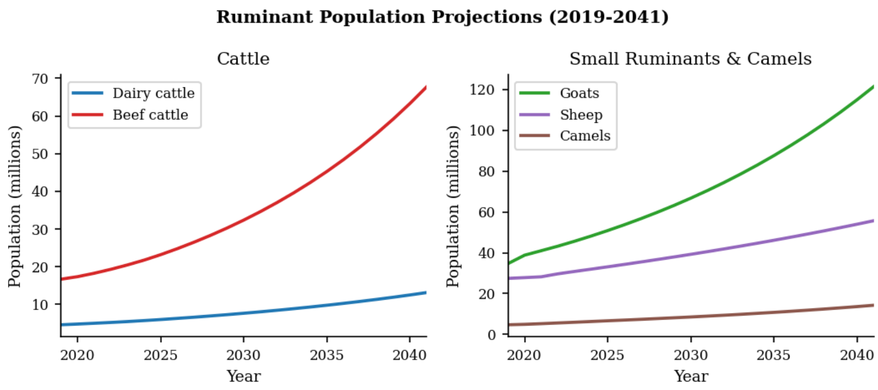

Fig. 1 Ruminant population projections, 2019–2041. Shaded region indicates the 2020–2025 validation window.

goats (156 million kg to 531 million kg; CAGR 5.71%). Chevon (goat meat) records the highest CAGR among all ruminant products at 17.02%, driven by the large baseline herd adjustment in 2020. Beef production grows from 322 million kg to 1.37 billion kg (CAGR 6.82%), and mutton from 49 million kg to 165 million kg (CAGR 5.65%). Wool production (sheep) is modest but grows consistently (CAGR 7.25%). Production trends are presented in Fig. 2.

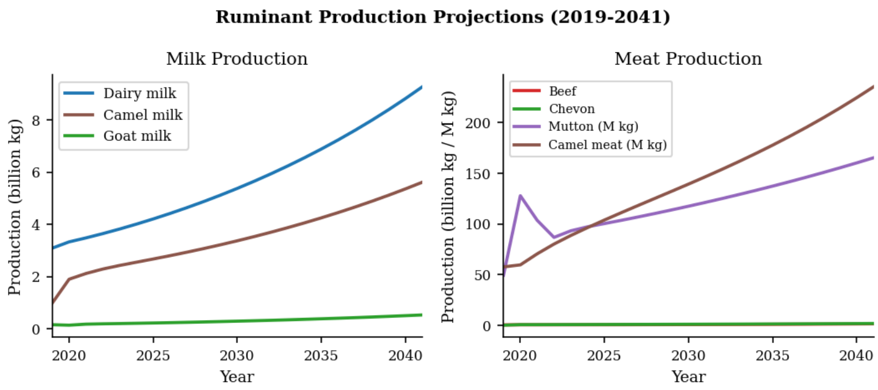

Fig. 2 Ruminant production projections, 2019–2041.

4.1.3 Feed Demand

Beef cattle are the dominant feed consumer throughout the horizon, rising from 57.4 million tonnes DM to 218.3 million tonnes DM (CAGR 6.26%). Dairy cattle demand grows from 14.3 million to 41.2 million tonnes DM (CAGR 4.92%). Camel feed demand expands at CAGR 6.66%, from 10.2 million to 42.2 million tonnes DM. Goat and sheep feed series begin in 2020 (no 2019 parameter baseline) and grow at CAGR

6.09% and 3.33% respectively over their available range. Feed demand trajectories are shown in Fig. 3.

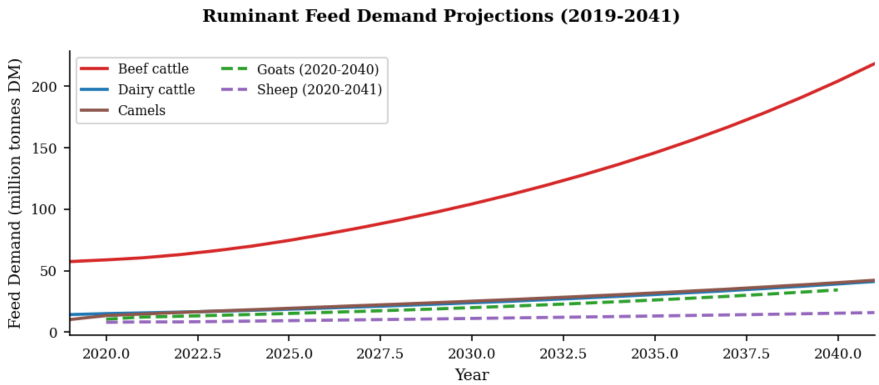

Fig. 3 Ruminant feed demand projections, 2019–2041. Goat and sheep series begin in 2020; the goat 2041 value is excluded due to a data entry anomaly.

4.2 Non-Ruminant Value Chains

4.2.1 Population

Indigenous poultry is the largest non-ruminant population, growing from 46.3 million to 89.3 million birds (CAGR 3.03%). Broiler throughput and layer flock size are relatively stable, growing at CAGR 0.43% and 0.42% respectively, reflecting sector saturation assumptions. Pig and rabbit populations expand more rapidly: pigs from 596 thousand to 2.09 million (CAGR 5.86%) and rabbits from 737 thousand to 3.24 million (CAGR 6.95%). Total apiculture hive count grows from 1.29 million to 3.49 million across all four hive types (log, KTBH, Langstroth, box). Population trends are shown in Fig. 4.

4.2.2 Production

Poultry meat production grows modestly from 82.4 million kg to 109.7 million kg (CAGR 1.31%), consistent with the near-stable flock projections. Egg production (layers and indigenous combined) is reported in trays from 2020 onward — the 2019 value is excluded as it appears to reflect a different unit convention — and grows from 180.4 million trays to 228.7 million trays (CAGR 1.14%, 2020–2041). Pork and rabbit meat grow in parallel at approximately CAGR 4.0%. Honey production expands from 13.9 million kg to 41.6 million kg (CAGR 5.12%) and beeswax from 1.4 million kg to 6.5 million kg (CAGR 7.35%). Production trends are shown in Fig. 5.

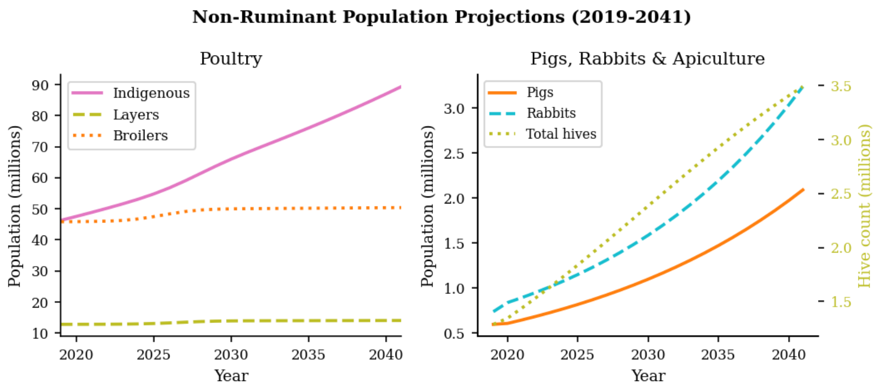

Fig. 4 Non-ruminant population projections, 2019–2041. Right panel: pig and rabbit populations (left axis) and total hive count (right axis).

4.2.3 Feed Demand

Poultry feed demand is the largest non-ruminant feed consumer, growing from 1.87 million tonnes to 2.83 million tonnes DM (CAGR 1.89%). Pig feed demand grows from 448 thousand tonnes to 1.09 million tonnes DM (CAGR 4.13%). Rabbit feed demand, though small in absolute terms (18 thousand to 53 thousand tonnes DM), grows at CAGR 4.94%. Apiculture nectar demand is estimated at 852 thousand tonnes in 2019 rising to 2.49 million tonnes by 2041 (CAGR 5.00%). Feed demand trajectories are shown in Fig. 6.

4.3 CAGR Summary

Table 1 summarises the CAGRs for all value chains across all three domains. All figures are 2019–2041 unless noted.

5 Discussion

Write discussion here.

6 Managerial, Policy, and Societal Implications

Write implications here.

7 Limitations and Opportunities for Future Research

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8 Conclusion

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Table 1 Compound Annual Growth Rates (2019–2041)

Value Chain	Indicator	CAGR (%)
Ruminants — Population		
Dairy cattle	Total population	4.93
Beef cattle	Total population	6.59
Goats	Total population	5.86
Sheep	Total population	3.27
Camels	Total population	5.15
Ruminants — Production		
Dairy cattle	Milk (kg)	5.12
Beef cattle	Beef (kg)	6.82
Goats	Milk (kg)	5.71
Goats	Chevon (kg)	17.02
Sheep	Mutton (kg)	5.65
Sheep	Wool (kg)	7.25
Camels	Milk (kg)	8.23
Camels	Meat (kg)	6.61
Ruminants — Feed Demand		
Dairy cattle	(t DM)	4.92
Beef cattle	(t DM)	6.26
Goats	(t DM, 2020–2040)	6.09
Sheep	(t DM, 2020–2041)	3.33
Camels	(t DM)	6.66
Non-Ruminants — Population		
Poultry (indigenous)	Birds	3.03
Poultry (layers)	Birds	0.42
Poultry (broilers)	Throughput	0.43
Pigs	Total	5.86
Rabbits	Total	6.95
Non-Ruminants — Production		
Poultry	Meat (kg)	1.31
Poultry	Eggs (trays, 2020–2041)	1.14
Pigs	Pork (kg)	3.99
Rabbits	Meat (kg)	4.01
Apiculture	Honey (kg)	5.12
Apiculture	Wax (kg)	7.35
Non-Ruminants — Feed Demand		
Poultry	(t DM)	1.89
Pigs	(t DM)	4.13
Rabbits	(t DM)	4.94
Apiculture	Nectar (t)	5.00

Fig. 5 Non-ruminant production projections, 2019–2041. Egg production plotted from 2020 onward only.

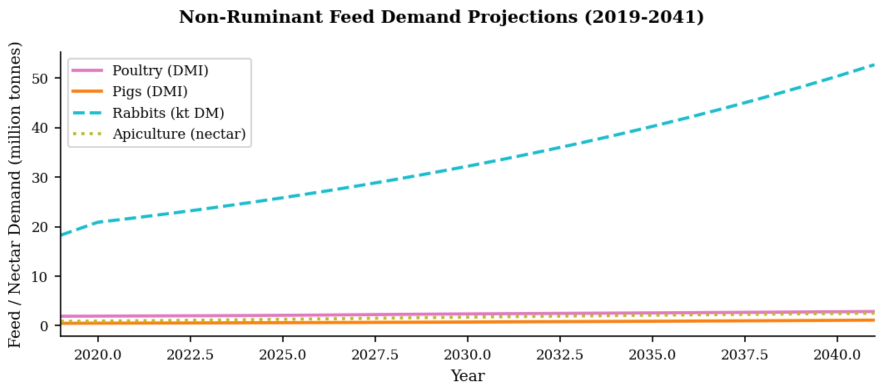

Fig. 6 Non-ruminant feed demand projections, 2019–2041. Apiculture values represent nectar demand rather than DMI.

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Declarations

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- Conflict of interest/Competing interests: Not applicable

- Ethics approval and consent to participate: Not applicable
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- Data availability: Not applicable
- Materials availability: Not applicable
- Code availability: Not applicable
- Author contribution: Not applicable

Appendix A Model Parameters

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